

INTERNATIONAL STANDARD



**Electricity metering data exchange –The DLMS/COSEM suite –
Part 8-5: Narrow-band OFDM G3-PLC communication profile
for neighbourhood networks**



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Part 8-5: Narrow-band OFDM G3-PLC communication profile
for neighbourhood networks**

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**ELECTRICITY METERING DATA EXCHANGE –
THE DLMS/COSEM SUITE –****Part 8-5: Narrow-band OFDM G3-PLC communication profile
for neighbourhood networks**

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International Standard IEC 62056-8-5 has been prepared by IEC technical committee 13: Electrical energy measurement and control.

The text of this International Standard is based on the following documents:

CDV	Report on voting
13/1708/CDV	13/1740/RVC

Full information on the voting for the approval of this International Standard can be found in the report on voting indicated in the above table.

A list of all parts in the IEC 62056 series, published under the general title *Electricity metering data exchange – The DLMS/COSEM suite*, can be found on the IEC website.

This document has been drafted in accordance with the ISO/IEC Directives, Part 2.

The committee has decided that the contents of this document will remain unchanged until the stability date indicated on the IEC website under "<http://webstore.iec.ch>" in the data related to the specific document. At this date, the document will be

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INTRODUCTION

As defined in IEC 62056-1-0, the IEC 62056 DLMS/COSEM suite provides specific communication profile standards for communication media relevant for smart metering.

Such communication profile standards specify how the COSEM data model and the DLMS/COSEM application layer can be used on the lower, communication media-specific protocol layers.

Communication profile standards refer to communication standards that are part of the IEC 62056 DLMS/COSEM suite or to any other open communication standard.

This International Standard specifies the DLMS/COSEM communication profile for ITU-T G.9903:2014 PLC communication based on OFDM technology.

ITU-T G.9903 PLC is designed to meet the following aims:

- Robustness: the communication profile shall be suited to severe powerline environments (see 5.3.2);
- Performance and scalability: it embeds adaptive modulation to use the proper modulation according to the quality of the link (see 5.3.2) within dense environments (up to 2 000 nodes in the same PAN);
- Security: it shall offer a secure environment (see 7.4);
- Openness: it shall be based on open standards in order to support multi-supplier solutions (see Clause 5);
- Flexibility and future proof: it shall be able to support future applications through using IPv6 networking capabilities (see 5.3.4).

This standard follows the rules defined in IEC 62056-5-3:2017, Annex A.

ELECTRICITY METERING DATA EXCHANGE – THE DLMS/COSEM SUITE –

Part 8-5: Narrow-band OFDM G3-PLC communication profile for neighbourhood networks

1 Scope

This part of IEC 62056 specifies the IEC 62056 DLMS/COSEM communication profile for metering purposes based on the Recommendations ITU-T G.9901: *Narrowband orthogonal frequency division multiplexing power line communication transceivers – Power spectral density specification* and ITU-T G.9903:2014, *Narrowband orthogonal frequency division multiplexing power line communication transceivers for G3-PLC networks*, an Orthogonal Frequency Division Multiplexing (OFDM) Power Line Communications (PLC) protocol.

The physical layer provides a modulation technique that efficiently utilizes the allowed bandwidth within the CENELEC A (3 kHz – 95 kHz), CENELEC B (95 kHz – 125 kHz), ARIB (10 kHz – 450 kHz) and FCC (no specific frequency band limitations) bands, thereby allowing the use of advanced channel coding techniques. This enables a robust communication in the presence of narrowband interference, impulsive noise, and frequency selective attenuation.

The medium access control (MAC) layer allows the transmission of MAC frames through the use of the power line physical channel. It provides data services, frame validation control, node association and secure services.

The 6LoWPAN adaptation sublayer enables an efficient interaction between the MAC and the IPv6 network layer. The use of the IPv6 network protocol – the latest generation of IP protocols – opens a wide range of potential applications and services for metering purposes (but the applications are not limited to metering).

The transport layer, the application layer and the data model are as specified in the IEC 62056 DLMS/COSEM suite.

The scope of this communication profile standard is restricted to aspects concerning the use of communication protocols in conjunction with the COSEM data model and the DLMS/COSEM application layer. Data structures specific to a communication protocol are out of the scope of this communication profile standard.

NOTE They are specified in the specific protocol standards.

Any project specific definitions of data structures and data contents may be provided in project specific companion specifications.

2 Normative references

The following documents are referred to in the text in such a way that some or all of their content constitutes requirements of this document. For dated references, only the edition cited applies. For undated references, the latest edition of the referenced document (including any amendments) applies.

IEC 60050-300, *International Electrotechnical Vocabulary – Electrical and electronic measurements and measuring instruments – Part 311: General terms relating to measurements – Part 312: General terms relating to electrical measurements – Part 313:*

Types of electrical measuring instruments – Part 314: Specific terms according to the type of instrument

IEC TR 62051, *Electricity metering – Glossary of terms*

IEC TR 62051-1, *Electricity metering – Data exchange for meter reading, tariff and load control – Glossary of terms – Part 1: Terms related to data exchange with metering equipment using DLMS/COSEM*

IEC 62056-1-0, *Electricity metering data exchange – The DLMS/COSEM suite – Part 1-0: Smart metering standardisation framework*

IEC 62056-4-7:2015, *Electricity metering data exchange – The DLMS/COSEM suite – Part 4-7: DLMS/COSEM transport layer for IP networks*

IEC 62056-5-3:2017, *Electricity metering data exchange – The DLMS/COSEM suite – Part 5-3: DLMS/COSEM application layer*

IEC 62056-6-1, *Electricity metering data exchange – The DLMS/COSEM suite – Part 6-1: Object identification system (OBIS)*

IEC 62056-6-2, *Electricity metering data exchange – The DLMS/COSEM suite – Part 6-2: COSEM interface classes*

IEC 62056-9-7:2013, *Electricity metering data exchange – The DLMS/COSEM suite – Part 9-7: Communication profile for TCP-UDP/IP networks*

Recommendation ITU-T G.9903:2014, *Narrowband Orthogonal Frequency Division Multiplexing Power Line Communication Transceivers for G3-PLC Networks available at <http://www.itu.int/rec/T-REC-G.9903/en>*

IETF RFC 768, *User Datagram Protocol. Edited by J. Postel. August 1980. Available from <http://www.ietf.org/rfc/rfc768.txt>*

IETF RFC 2460, *Internet Protocol, Version 6 (IPv6) Specification. Edited by S. Deering, R. Hinden. December 1998. Available from <http://www.ietf.org/rfc/rfc2460.txt>*

IETF RFC 4193, *Unique Local IPv6 Unicast Addresses. Edited by R. Hinden, B. Haberman. October 2005. Available from <http://www.ietf.org/rfc/rfc4193.txt>*

IETF RFC 4291, *IP Version 6 Addressing Architecture. Edited by R. Hinden, S. Deering. February 2006. Available from <http://www.ietf.org/rfc/rfc4291.txt>*

IETF RFC 4944, *Transmission of IPv6 Packets over IEEE 802.15.4 Networks. Available from <http://www.ietf.org/rfc/rfc2460.txt>*

IETF RFC 6282, *Compression Format for IPv6 Datagrams over IEEE 802.15.4-Based Networks. Available from <http://www.ietf.org/rfc/rfc2460.txt>*

IETF RFC 4861, *Neighbor Discovery for IP version 6 (IPv6). Available from <http://www.ietf.org/rfc/rfc4861.txt>*

IETF RFC 4862, *IPv6 Stateless Address Autoconfiguration. Available from <http://www.ietf.org/rfc/rfc4862.txt>*

IEEE 802.15.4: *IEEE Standard for Low-Rate Wireless Networks*

3 Terms, definitions and abbreviated terms

3.1 Terms and definitions

For the purposes of this document, the terms and definitions given in IEC 60050-300, IEC TR 62051, IEC TR 62051-1 and the following apply.

NOTE Where there is a difference between the definitions in the glossaries and those contained in communication profile standards established by TC 13, then the latter take precedence in applications of the relevant standard.

ISO and IEC maintain terminological databases for use in standardization at the following addresses:

- IEC Electropedia: available at <http://www.electropedia.org/>
- ISO Online browsing platform: available at <http://www.iso.org/obp>

3.1.1

PAN coordinator

entity implementing the G3-PLC protocol capable of controlling the network

3.1.2

PAN device

entity implementing the G3-PLC protocol which does not embed coordinator functionalities

3.2 Abbreviated terms

AA	Application Association
APDU	Application Layer Protocol Data Unit
ARIB	Association of Radio Industries and Businesses (Japan)
6LoWPAN	IPv6 over Low power Wireless Personal Area Networks
CENELEC	European Committee for Electrotechnical Standardization
COSEM	Companion Specification for Energy Metering
DLMS	Device Language Message Specification
FCC	Federal Communications Commission (US)
IEC	International Electrotechnical Commission
IP	Internet Protocol
ITU-T	International Telecommunication Union – Telecommunication
LBA	LoWPAN Bootstrapping Agent
LBP	LoWPAN Bootstrapping Protocol
MAC	Media Access Control
NNAP	Neighbourhood Network Access Point
OFDM	Orthogonal Frequency Division Multiplexing
OSI	Open System Interconnection
PAN	Personal Area Network
PLC	Power Line Communication
PSK	Pre-Shared Key
TCP	Transmission Control Protocol
UDP	User Datagram Protocol

Furthermore, the abbreviations given in ITU-T G.9903:2014, Clause 4 also apply.

4 Targeted communication environments

The DLMS/COSEM narrow-band OFDM G3-PLC communication profile for neighbourhood networks is intended for remote data exchange on Neighbourhood Networks (NN) between Neighbourhood Network Access Points (NNAPs) and Local Network Access Points (LNAPs) or End Devices using OFDM technology over the low voltage electricity distribution network as a communication medium. The functional reference architecture is shown in Figure 1.

End devices – typically electricity meters – comprise application functions and communication functions. They may be connected directly to the NNAP via the C interface, or to an LNAP via an M interface, while the LNAP is connected to the NNAP via the C interface. The LNAP function may be co-located with the metering functions.

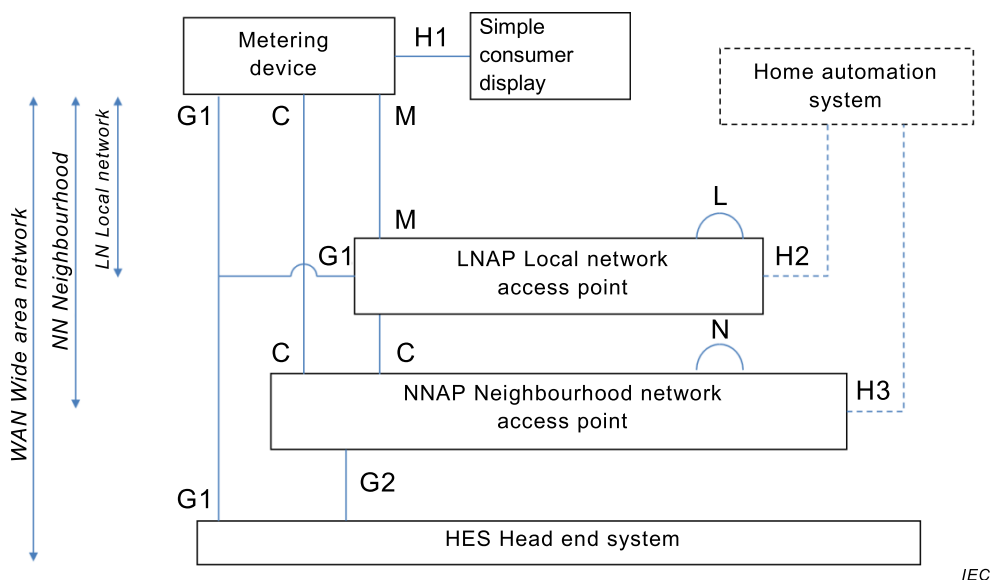
A NNAP comprises gateway functions and it may comprise concentrator functions. Upstream, it is connected to the Metering Head End System (HES) using suitable communication media and protocols. The communication channel between the NNAP and HES is out of scope of this document.

End devices and LNAPs may communicate to different NNAPs, but to one NNAP only at a time. From the PLC communication point of view, the NNAP acts as the PAN coordinator while end devices and LNAPs act as PAN devices.

NNAPs and similarly LNAPs may communicate to each other, but this is out of the scope of this document, which covers the C interface only.

When the NNAP has concentrator functions, it acts as a DLMS/COSEM client. When the NNAP has gateway functionality only, then the HES plays the role of a DLMS/COSEM client. The end devices or the LNAPs play the role of DLMS/COSEM servers.

A mixed architecture is also possible, i.e. both the HES and the NNAP can act as a client.



IEC

Figure 1 – Entities and interfaces of a smart metering system using the terminology of IEC 62056-1-0

5 Use of the communication layers for this profile

5.1 Information related to the use of the standard specifying the lower layers

The proposed communication model natively integrates a network layer and an IP suite transport layer which opens the way to a vast range of Internet applications and ensures great flexibility in the system architecture. It provides the possibility of having:

- either a decentralized architecture, where the NNAP (also known as data concentrator) acts as an application relay, with more or less autonomy. The exchanges at application level in this case are limited to the dialogue between the meters / LNAPs and the NNAPs;
- or a more centralized architecture in which the NNAP simply acts as a network gateway and the meters (LNAPs) dialogue directly with HES from an application layer point of view).

A combination of these architectures is also possible. A centralized architecture can be dedicated for sensitive and/or on-demand requests whereas distributed architecture can be used for recurrent operations.

The network layer chosen is based on the IPv6 protocol (IETF RFC 2460).

The protocol used for the transport layer is UDP (IETF RFC 768), which provides unreliable transport of datagrams in connectionless mode. Reliability of exchanges within the PLC network is brought by the combination of lower layer robustness (such as retries or physical layer forward error correction) and DLMS/COSEM services.

The ITU-T G.9903:2014 specification defines a standardized header compression mechanism based on IETF RFC 6282 enabling IPv6 and UDP header compression.

5.2 Structure of the communication profiles

The protocol stack uses the following protocol layers as shown in Figure 2:

- The DLMS/COSEM Application layer as specified in IEC 62056-5-3 covering the Application, Presentation and Session functionalities;
- The DLMS/COSEM Transport layer as specified in IEC 62056-4-7:2015, used with the DLMS/COSEM UDP/IPv6 profile over the G3-PLC network;
- The ITU-T G.9903:2014 Data link layer, that consists of the IETF 6LoWPAN Adaptation Layer (IETF RFC 4944, IETF RFC 6282) and the MAC sublayer;
- The ITU-T G.9903:2014 Physical layer adapted to the band used (see ITU-T G.9903:2014, Clause 7).

Following this reference model, a profile is fully defined using G3-PLC physical and data link layers, UDP/IPv6, the DLMS/COSEM application layer and the COSEM object model.

NOTE The COSEM interface classes for setting up and managing data exchange over the narrow-band OFDM G3-PLC networks are specified in IEC 62056-6-2.

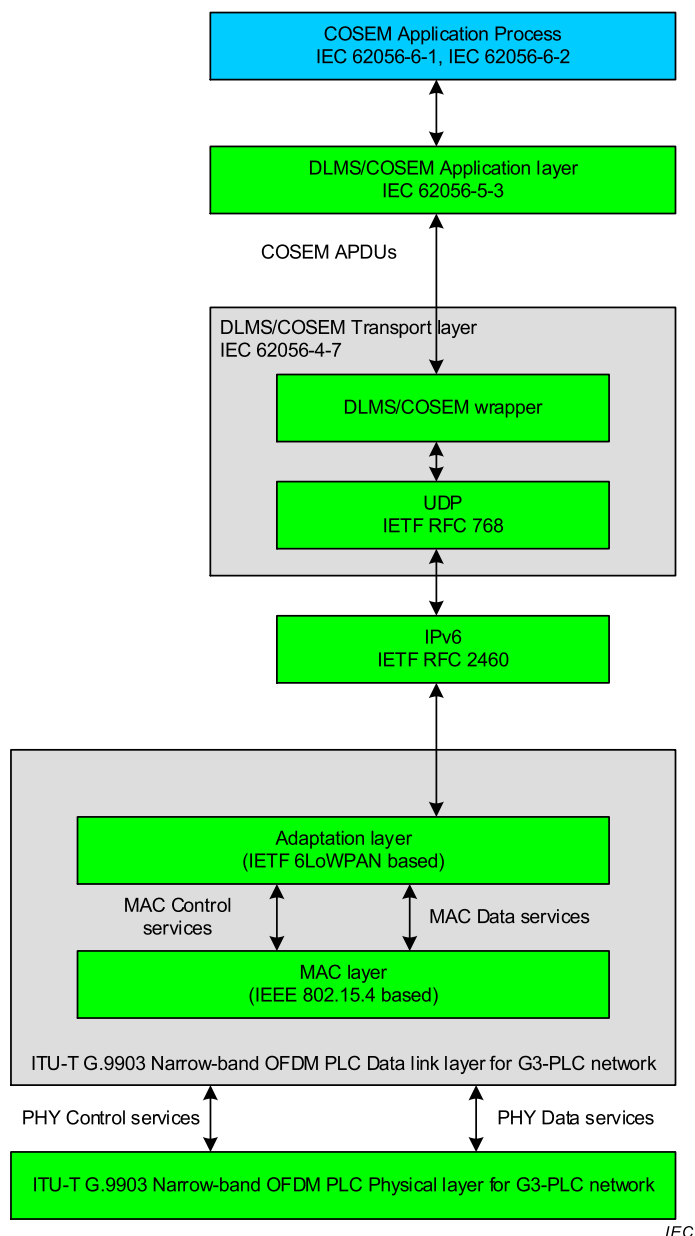


Figure 2 – G3-PLC protocol architecture

5.3 Lower protocol layers and their use

5.3.1 Overview

The UDP/IPv6 based communication profile is fully in line with the DLMS/COSEM communication profile for UDP/IP, as specified in IEC 62056-9-7:2013. Refer to that standard for more information. This subclause provides information related to the binding of UDP/IPv6 layers with the ITU-T G.9903:2014 protocol layers.

The general architecture is the following, see Figures 3 and 4:

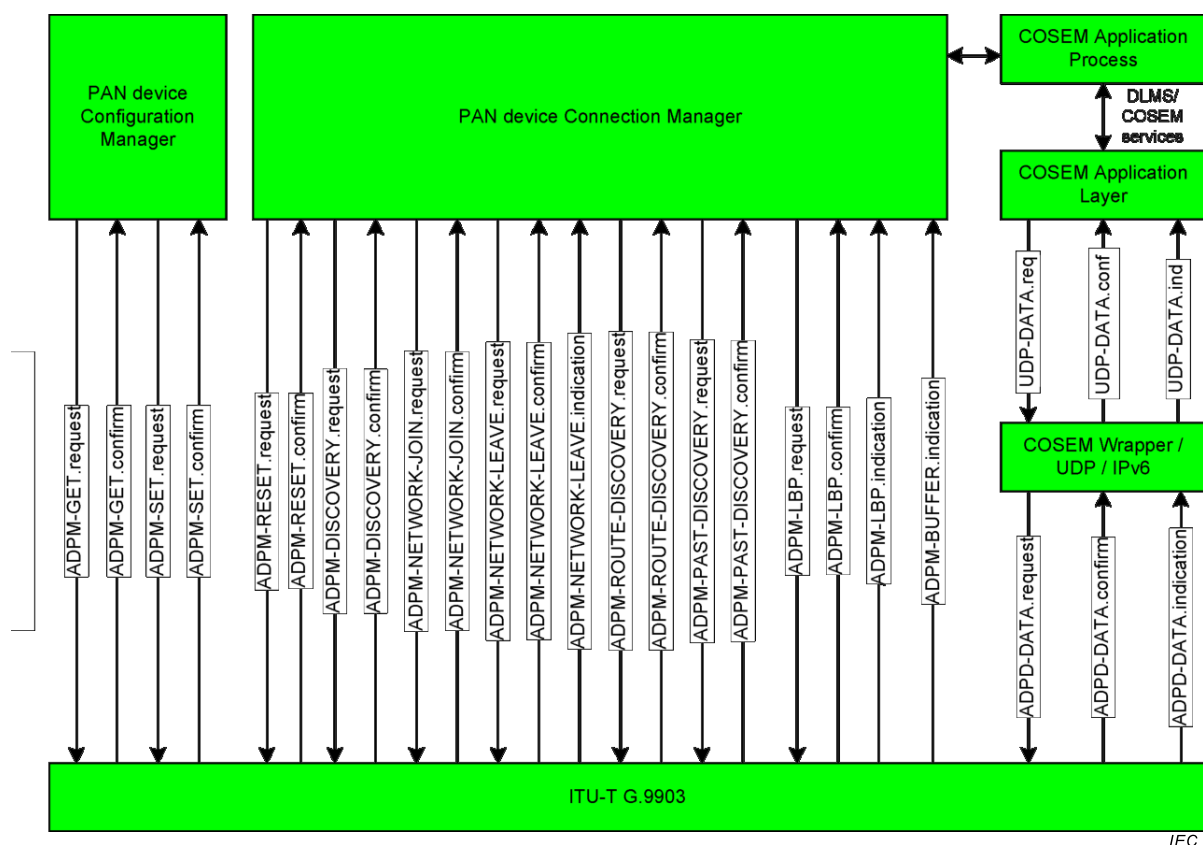


Figure 3 – PAN device communication profile architecture

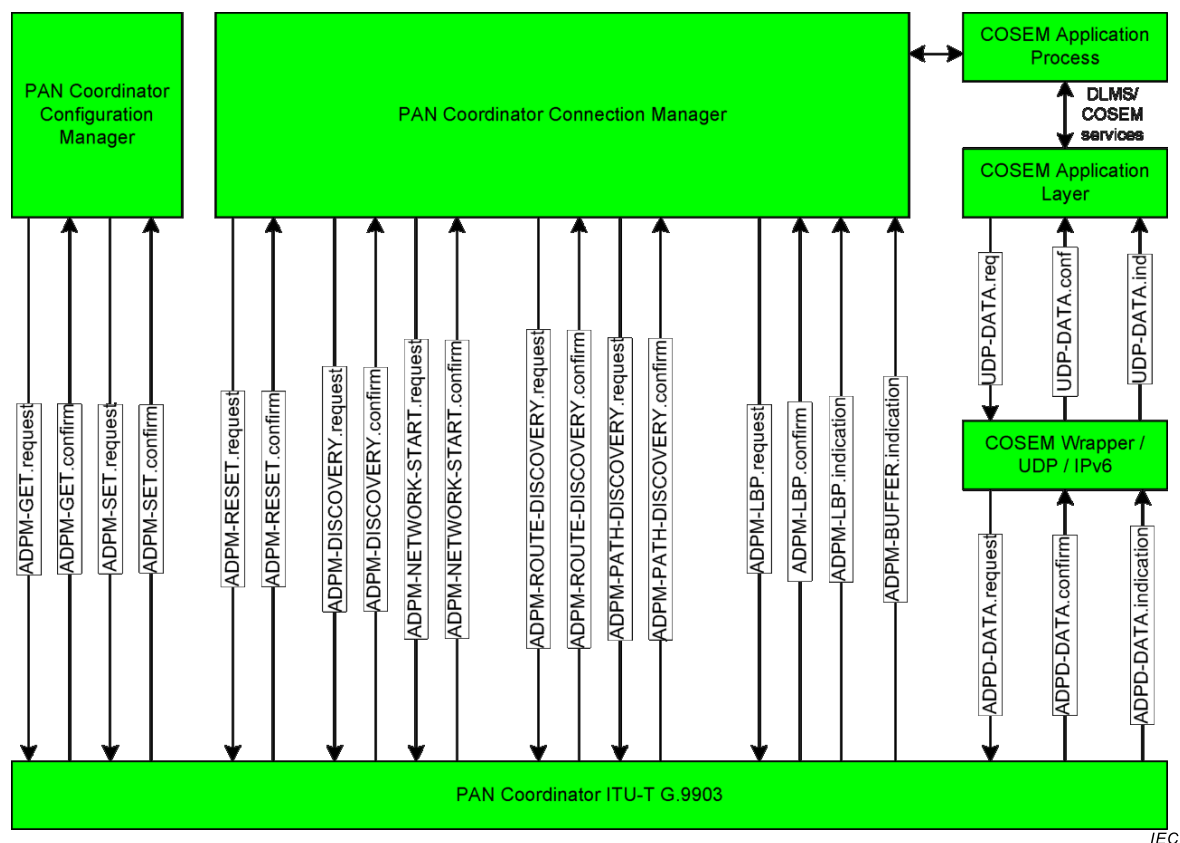


Figure 4 – PAN coordinator communication profile architecture

The Configuration Manager (for the PAN device and the PAN Coordinator) is in charge of managing the configuration of the ITU-T G.9903:2014 sublayers. The following service primitives are used to access ITU-T G.9903:2014 configuration attributes:

- ADPM-SET.request, .confirm;
- ADPM-GET.request, .confirm.

For more details about configuration service primitives, see ITU-T G.9903:2014, 9.4.6.

5.3.2 Physical layer

5.3.2.1 Overview

This layer provides the interface between the equipment and the physical transmission medium that is the power line. It transmits and receives MPDUs between neighbour nodes. The physical layer uses OFDM modulation. The OFDM signal uses a frequency bandwidth depending on the frequency band used: 54,6875 kHz for the CENELEC A band for example.

For more details about the G3-PLC PHY layer, see ITU-T G.9903:2014, Clause 7.

5.3.2.2 G3-PLC PHY data plane services

G3-PLC PHY data plane services are generated / used by the MAC layer entity whenever data have to be transmitted to / received from (a) peer MAC entity(ies) using the PHY transmission procedures. See ITU-T G.9903:2014, 7.17.1. They are the following:

- PD-DATA.request: allows the MAC layer entity to request the transmission of an MPDU to a peer MAC entity;
- PD-DATA.confirm: allows the PHY layer entity to confirm the end of the transmission of an MPDU to the local MAC entity;
- PD-DATA.indication: allows the PHY layer entity to transfer a PPDU from the PHY to the local MAC entity;
- PD-ACK.request: allows the MAC layer entity to send an ACK PHY frame to the peer MAC entity via the local PHY entity;
- PD-ACK.confirm: allows the PHY layer entity to confirm the end of the transmission of an ACK PHY frame to the local MAC entity;
- PD-ACK.indication: allows the PHY layer entity to indicate that an ACK PHY frame has been received from a peer MAC entity.

5.3.2.3 G3-PLC PHY management plane services

G3-PLC PHY management plane services are used to manage the physical layer by the MAC layer. See ITU-T G.9903:2014, 7.17.2. They are described below:

- PLME_SET: allows the MAC layer entity to set the characteristics (power, modulation, etc.) to be used for PHY packet transmission;
- PLME_GET: allows the MAC layer entity to get physical information from the last physical packet received: channel characteristics (power, modulation, etc.), the signal noise ratio (SNR) global or per each carrier and the phase differential with the neighbour;
- PLME_SET_TRX_STATE: allows the MAC layer entity to change the mode of the PHY layer either in transmission or reception mode;
- PLME_CS: allows the MAC layer entity to get media status using carrier sense (either idle or busy). This information is used by the CSMA-CA algorithm implemented in the MAC layer.

5.3.3 MAC layer

5.3.3.1 Overview

A G3-PLC subnetwork is a mesh network with two types of nodes, PAN Coordinator and PAN device. The PAN coordinator is at the root of the mesh network and acts as the master node that provides the subnetwork with connectivity. There is one PAN coordinator in a subnetwork. Any other subnetwork node is a PAN device.

The PAN Coordinator manages the G3-PLC subnetwork's resources and connections. PAN devices have to follow a registration process to securely join the subnetwork.

PAN devices start in a "non-associated" state before discovering their environment (neighbouring PAN coordinators and PAN devices) to register to the G3-PLC subnetwork. Once the registration process has been successfully completed, the PAN device receives a short MAC address and may automatically serve as a router for other PAN devices attempting to connect to the subnetwork.

The functions offered by the MAC layer are:

- Channel Access implementation using the CSMA/CA algorithm;
- Inter-frame (IFS) spacing management;
- Management of the priority access to the channel following the type of MPDU to transmit;
- Automatic Repeat Request (ARQ) mechanism between nodes based on acknowledged and unacknowledged retransmission;
- MAC packet (MPDU) segmentation and reassembly to / from PHY layer;
- MAC packet (MPDU) ciphering.

5.3.3.2 G3-PLC MAC data services

G3-PLC MAC data services are generated / used by the Adaptation layer entity whenever data have to be transmitted to / received from (a) peer Adaptation entity(ies) using the MAC transmission procedures. See ITU-T G.9903:2014, 9.3.3. They are the following:

- MCPS-DATA: is used for sending or receiving unicast, multicast or broadcast data.

5.3.3.3 G3-PLC MAC management services

G3-PLC MAC management services are generated / used between the local Adaptation layer entity and the local MAC entity. See ITU-T G.9903:2014, 9.3.3. They are the following:

- MLME-BEACON-NOTIFY: allows the MAC layer entity to send to the Adaptation layer entity information received from a neighbour during neighbourhood discovery;
- MLME-GET: allows the Adaptation layer entity to request the value of a given configuration PIB attribute from the MAC layer entity;
- MLME-RESET: allows the Adaptation layer entity to reset the MAC layer entity to its default state;
- MLME-SCAN: used by the Adaptation layer entity to initiate an active channel scan to discover the neighbourhood G3-PLC network;
- MLME-COMM-STATUS: allows the local MAC layer entity to inform the Adaptation layer entity about the status of a communication;
- MLME-SET: allows the Adaptation layer entity to set the value of a given configuration PIB attribute of the MAC layer entity;
- MLME-START: allows the Adaptation layer entity to initiate a new G3-PLC subnetwork (new PAN). This service is only used by the PAN Coordinator;

- MLME-SYNC-LOSS: allows the MAC layer entity to indicate to the Adaptation layer entity a detection of an alternate PAN network (i.e. another G3-PLC subnetwork managed by another PAN Coordinator).

5.3.4 Network layer – IPv6

5.3.4.1 Overview

This subclause describes how IPv6 is supported over ITU-T G.9903 layers.

The IPv6 protocol has to be implemented in conformance with IETF RFC 2460 and the appropriate updates to the protocol. The following subclauses specify the IPv6 addressing plan meeting architectural considerations, guaranteeing overall scalability and enabling IP end-to-end communications (direct IP communication between the HES and the meters as mentioned in 5.1).

5.3.4.2 IPv6 addressing plan

The most important feature of IPv6 is a much larger address space than that of IPv4: IPv6 addresses length is 128 bits compared to the 32-bit IPv4 addresses. Furthermore, compared to IPv4, IPv6 supports multi-addressing on one physical interface (global, unique or link local IPv6 addresses).

IPv6 addresses are typically composed of two logical parts: a 64-bit (sub-)network prefix used for routing, and a 64-bit host part used to identify a host within the network.

The formats allowed for an IPv6 address are shown in Figure 5 (see <http://www.iana.org/assignments/ipv6-address-space/>). In order to facilitate the representation of IPv6 addresses, a specific notation defined in IETF RFC 4291 has been specified by IETF (e.g. FF00::/8).

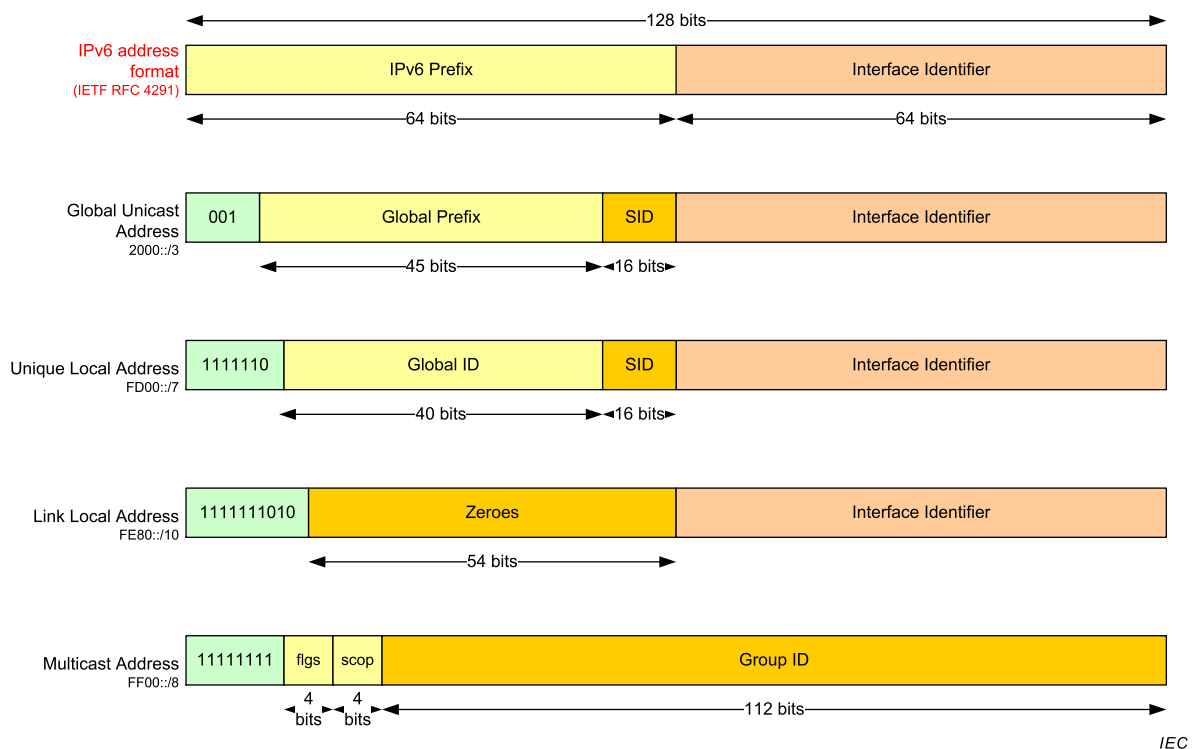


Figure 5 – IPv6 address formats

Where:

- Global Unicast Address (GUA) is a routable address in the whole internet network and is composed as follows:
 - Global Prefix assigned by IANA (see <http://www.iana.org/assignments/ipv6-unicast-address-assignments/>);
 - Subnet ID (SID) allocated by the network administrator; and
 - Interface Identifier either generated from the interface's MAC address (using modified EUI-64 format), or obtained from a DHCPv6 server, or assigned manually;
- Unique Local Address (ULA) defined in IETF RFC 4193 is a unicast address only applicable to a private network. This type of address is not routable outside the private network. The Global ID and the Subnet ID (SID) are allocated by the network administrator;
- Link Local Address is a unicast address allowed for a local link. This type of address is not routable outside a local link;
- Multicast Address allows simultaneous addressing of different devices of a network. Following the scope (scop field) of the address, the multicast group may be either Interface-local, Link-local, Admin-local, Site-local, Organization-local or global. For more information about the flgs and scop fields, see IETF RFC 4291:2006, 2.7.

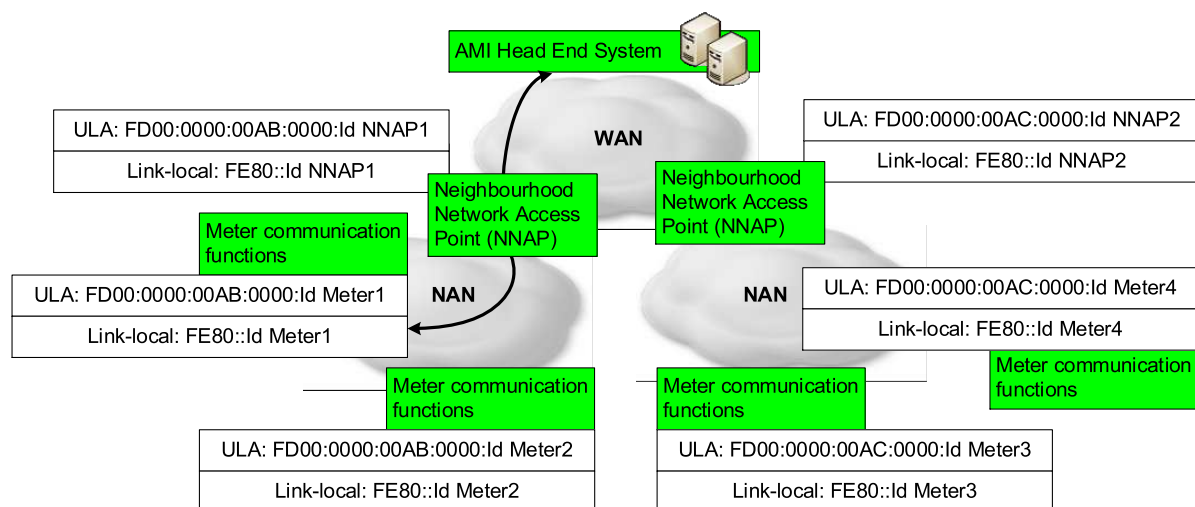
It is important to note that there is no broadcast address defined in IPv6.

For more information concerning IPv6 addressing, see IETF RFC 4291.

ITU-T G.9903:2014 allows the use of all these address types. It is to be noted that both PAN coordinators and PAN devices are provisioned with one default link-local scope multicast address (0xFF02::0001) corresponding to the 0x8001 adaptation layer group ID.

ULA type addresses are recommended to be used for end-to-end communication between external IP hosts and devices belonging to the subnetwork in addition to the link-local addresses the use of which is restricted to local communication between devices belonging to the subnetwork. Alternatively, GUA addresses may be used as well (although the ability to route this type of addresses over the Internet should be taken into account).

Figure 6 shows an example of IPv6 addressing plan using the ITU-T G.9903:2014 communication profile:



IEC

Figure 6 – IPv6 Addressing plan example

Prefix construction is based on the 0xFD/8 static ULA prefix, the global ID is used to guarantee a unique prefix for each NNAP (in the example given, the Global IDs 0x00:0000:00AB and 0x00:0000:00AC are used) and the Subnet ID are reserved for future use (it is set to 0x0000).

For each PAN device in the G3-PLC network, the COSEM configuration object “IPv6Setup” (class_id = 48, see IEC 62056-6-2) lists the valid multicast addresses which reflects the possible combinations derived from the group table entries and, eventually, relevant context information.

5.3.4.3 IPv6 address provisioning

In order to comply with the IPv6 addressing plan outlined previously, IPv6 address provisioning is carried out in two steps:

- When a new PAN device joins a G3-PLC PAN, the bootstrapping procedure specified in ITU-T G.9903:2014 is applied as stated in ITU-T G.9903:2014, 9.4.4.2.2. This commissioning procedure results in the secure delivery of an IPv6 link-local address composed as follows; see Figure 7.

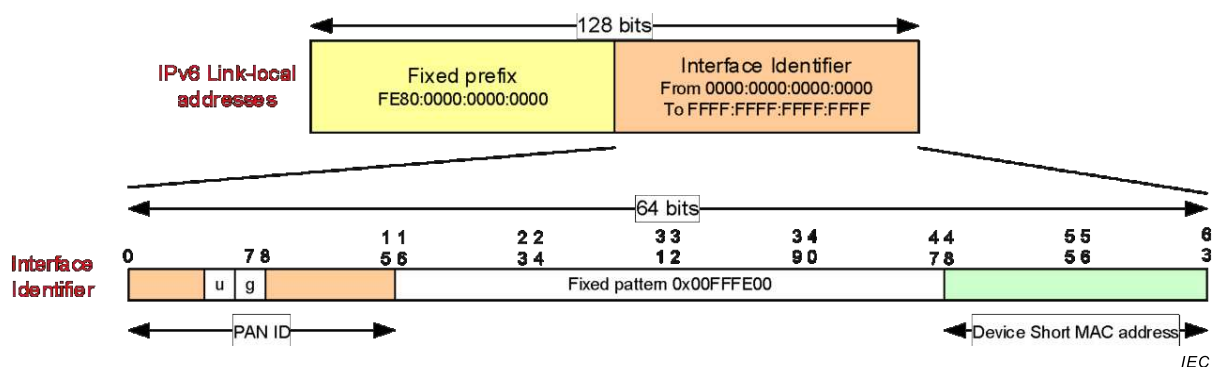


Figure 7 – IPv6 Link-local address composition

- The link-local prefix (FE80::/10) and a 64-bit interface identifier derived from the 16-bit PAN ID; and
- The 16-bit short address allocated by the PAN Coordinator during the bootstrapping procedure.

The 16-bit short addresses shall be provisioned as described in Table 1:

Table 1 – 16-bit short addresses allocation rule

Device type	Valid range
PAN Coordinator	0x0000
PAN device	0x0001 – 0x7FFF
Multicast addresses	0x8000 – 0x9FFF
Reserved addresses	0xA000 – 0xFFFE

- Once the PAN device has successfully joined the network, a second phase consists in the assignment of an IPv6 ULA address to the device, composed of a prefix advertised by the PAN coordinator and a 64-bit interface identifier corresponding to the interface identifier of the previously provisioned IPv6 link-local address. For the management of the IPv6 prefix, see ITU-T G.9903:2014, Table 9-28.

5.3.5 Transport layer – UDP

This subclause focuses on the use of the UDP protocol (see IETF RFC 768) within the ITU-T G.9903:2014 communication profile. The DLMS/COSEM Application layer has an interface with the UDP transport layer through the UDP/COSEM wrapper.

Table 2 summarizes the valid UDP port numbers for the DLMS/COSEM over the ITU-T G.9903:2014 profile:

Table 2 – UDP port numbering

Application	DLMS Server UDP ports		DLMS Client UDP ports	
	Source	Destination	Source	Destination
DLMS/COSEM	61616	61617 – 61629	61617 – 61629	61616

This choice has been made to ensure the best performance of the UDP/IPv6 header compression algorithm embedded in the ITU-T G.9903:2014 Adaptation layer (see ITU-T G.9903:2014, 9.4.2.2).

5.4 Service mapping and adaptation layers

5.4.1 Overview

NOTE The following description is based on ITU-T G.9903:2014, 9.4.

The functions offered by the IPv6 Adaptation layer are:

- Bootstrapping and authentication process;
- UDP and IPv6 header compression;
- LOADng Mesh Routing protocol (may be disabled);
- IPv6 Address auto-configuration;
- Fragmentation and reassembly of IPv6 packets;
- G3-PLC subnetwork management;
- Unicast and multicast.

5.4.2 G3-PLC Adaptation data services

G3-PLC Adaptation data plane services are generated / used by the application layer whenever data has to be transmitted to / received from (a) peer(s) using the Adaptation transmission procedures. See ITU-T G.9903:2014, 9.4.6.1. It is the following:

- ADPD-DATA: is used to send or receiving unicast, multicast or broadcast data.

5.4.3 G3-PLC Adaptation management services

G3-PLC Adaptation management plane services are generated / used by the local application, layer entity to the local Adaptation layer entity. See ITU-T G.9903:2014, 9.4.6.2. They are the following:

- ADPM-DISCOVERY: allows the upper layers to request to the Adaptation layer to scan for G3-PLC subnetwork in its POS;
- ADPM-NETWORK-START: allows the upper layers to request the starting of a new G3-PLC subnetwork. It shall only be invoked by a device designated as the PAN Coordinator of the new G3-PLC subnetwork;
- ADPM-NETWORK-JOIN: for a PAN device, allows the upper layers to request to register to a specific G3-PLC network (identified by its PAN identifier). On the other hand, for the

PAN Coordinator, this service primitive is used to inform the upper layers about a new request of registering;

- ADPM-NETWORK-LEAVE: allows the upper layers of a PAN device to remove itself from the G3-PLC subnetwork. This service primitive is not applicable for the PAN Coordinator;
- ADPM-RESET: allows the upper layers to request a reset of the G3-PLC PAN device or PAN coordinator (including the Adaptation and MAC layers);
- ADPM-GET: allows the upper layers to request the value of a given configuration PIB attribute of the Adaptation and MAC layer entities;
- ADPM-SET: allows the upper layers to set the value of a given configuration PIB attribute of the Adaptation and MAC layer entities;
- ADPM-NETWORK-STATUS: allows the upper layers of a PAN Coordinator to be notified when the Adaptation layer has received an LBP message from a device on the network indicating that a PAN Id conflict is occurring;
- ADPM-ROUTE-DISCOVERY: allows the upper layers to obtain a route to a given G3-PLC device;
- ADPM-PATH-DISCOVERY: collects the Link Costs all along the forward and reverse route to a given G3-PLC device;
- ADPM-LBP: allows the upper layers of a PAN device to send the LBP message to the PAN Coordinator. The LBP messages are used during the registration (bootstrapping procedure). This service primitive is not applicable to the PAN Coordinator;
- ADPM-Buffer: allows the adaptation layer to inform the upper layers about the fact that the modem has reached its capacity limit to process the next frame.

5.5 Registration and connection management

5.5.1 PAN device Connection Manager

5.5.1.1 Overview

The PAN device Connection Manager is in charge of:

- Managing the bootstrapping procedure (registration procedure to an established G3-PLC network);
- Leaving the associated G3-PLC subnetwork if requested by the application layer;
- Managing the G3-PLC communication route for diagnostic / maintenance and performance purposes.

The purpose of the following subclauses is to describe the different procedures listed above. Clauses A.1 and A.2 illustrate the G3-PLC PAN management procedures.

5.5.1.2 Bootstrapping procedure

The bootstrapping procedure defines the start-up procedure executed by any PAN device to register itself to a G3-PLC network. This procedure is divided into several steps as follows:

- a) Neighbourhood discovering phase: permits to discover the different PAN devices or PAN Coordinator which are in the local POS (direct connection without using router to communicate). Associated service primitives used: ADPM-DISCOVERY.request, .confirm;
- b) Access control phase: based on the information of the discovery, the PAN device Connection Manager is able to select a LBA (which is a G3-PLC node that has already joined the PAN and thus, it is already a member of the PAN) to send a registration request to the PAN Coordinator managing the selected G3-PLC network. Associated service primitive used: ADPM-NETWORK-JOIN.request;
- c) Authentication and key distribution phase: permits to have a mutual authentication and initial exchange of ciphering material with the PAN Coordinator. Also, this procedure permits a mutual control of session keys derivation;

- d) Authorization and initial configuration phase: once the authentication is successfully done, the PAN Coordinator assigns to this new PAN device a 16-bit short MAC address. Associated service primitive used: ADPM-NETWORK-JOIN.confirm.

At the end of the bootstrapping procedure, the PAN device is registered to the PAN Coordinator and may exchange IPv6 packets by using an optimized communication route.

For more detail about this procedure, see ITU-T G.9903:2014, 9.4.4.2.2.

5.5.1.3 Leaving a PAN

A PAN device may leave a PAN in two ways:

- a) Either by requesting it to the PAN Coordinator. For more detail, see ITU-T G.9903:2014, 9.4.4.2.2.8. Associated service primitives used:
- ADPM-NETWORK-LEAVE.request, .confirm.
- b) Or by receiving an order from the PAN Coordinator. For more detail, see ITU-T G.9903:2014, 9.4.4.2.2.7. Associated service primitive used:
- ADPM-NETWORK-LEAVE.indication.

At the end of this process, the 16-bit short address allocated previously by the PAN Coordinator shall be set to 0xFFFF. The PAN device cannot exchange data packets within the PAN anymore, unless it starts a new bootstrapping procedure as described in ITU-T G.9903:2014, 9.4.4.

5.5.1.4 Managing the G3-PLC network

The Recommendation ITU-T G.9903:2014 supports a mesh routing algorithm embedded in the 6LoWPAN adaptation sub-layer: LOADng (Lightweight On-demand Ad hoc Distance-vector routing protocol – next generation). This reactive routing algorithm is able to establish upstream and downstream routes from a node to any other node in the PAN in an efficient manner. It is also capable of handling unidirectional links and to find alternative routes if the intermediate link is broken or if the intermediate device runs out of memory. By invoking specific services, the PAN device Connection Manager may request:

- Manual route discovery: this can be triggered for maintenance or performance purposes. This mechanism aims at forcing the installation of a route even if no data has to be sent over the PAN. For more details, see ITU-T G.9903:2014, 9.4.3.2.3. Associated service primitives used: ADPM-ROUTE-DISCOVERY.request, confirm;
- Path discovery: this can be triggered for maintenance or performance purposes. The goal is to retrieve characteristics (quality, modulation, tone maps, etc.) of each link constituting the forward and reverse route to a specified destination node. For more details, see - ADPM PATH-DISCOVERY.request, .confirm.

5.5.2 PAN Coordinator Connection Manager

5.5.2.1 Overview

The PAN Coordinator Connection Manager is in charge of:

- Initialisation of the G3-PLC network;
- Managing the association requests from PAN devices;
- Leaving a PAN – removal of a device by the PAN coordinator;
- Managing the G3-PLC network.

The purpose of the following subclauses is to describe the different procedures listed above.

5.5.2.2 Initialisation of the G3-PLC network

To start the G3-PLC network, the PAN Coordinator Connection Manager shall initiate its own PAN to permit PAN devices in the neighbourhood to register / associate to the new G3-PLC subnetwork. The procedure is fully defined in ITU-T G.9903:2014, Annex C. Associated service primitives used:

- ADPM-RESET.request, .confirm;
- ADPM-DISCOVERY.request, .confirm;
- ADPM-NETWORK-START.request, .confirm.

5.5.2.3 Managing the association requests from PAN devices

The main task of the PAN Coordinator Connection Manager is to manage the joining request of PAN devices. This procedure is divided into several steps as follow:

- a) Access control phase: on reception of a new registration request, the PAN Coordinator shall compare the EUI-64 address of the new PAN device with an access control list (white or black list). If the address is authorized, the bootstrapping procedure may continue. Otherwise, the request shall be declined by the PAN Coordinator. The implementation of the access control list is out of scope of this specification. For more details, see ITU-T G.9903:2014, 9.4.4.2.2.3. Associated service primitives used:
 - ADPM-LBP.request, .confirm, .indication.
- b) Authentication and key distribution phase: permits to have a mutual authentication and initial exchange of ciphering material with the PAN device. Also, this procedure permits a mutual control of session key derivation. The management of the key is out of scope of this specification. For more details, see ITU-T G.9903, 9.4.4.2.2.4. Associated service primitives used:
 - ADPM-LBP.request, .confirm, .indication.
- c) Authorization and initial configuration phase: once the new PAN device is authenticated, the PAN Coordinator selects a 16-bit short address, globally defined and fully routable in the PAN and sends it back to the PAN device. It is recommended that one short address is assigned to one EUI-64 address for a given Group session key. For more details, see ITU-T G.9903:2014, 9.4.4.2.2.5. Associated service primitives used:
 - ADPM-LBP.request, .confirm, .indication.

For more detail about this procedure, see ITU-T G.9903:2014, 9.4.4.2.2.

5.5.2.4 Leaving a PAN – removal of a device by the PAN coordinator

The PAN coordinator may instruct a device to remove itself from the network invoking the ADPM-LBP.request service primitive, using a KICK frame. Associated service primitives used:

- ADPM-LBP.request, .confirm, .indication,

For more details, see ITU-T G.9903:2014, 9.4.4.2.2.7.

5.5.2.5 Managing the G3-PLC network

By invoking specific services, the PAN Coordinator Connection Manager may request:

- Manual route discovery: this can be triggered for diagnosis. For more detail, see ITU-T G.9903:2014, 9.4.3.2.3. Associated service primitives used: ADPM-ROUTE-DISCOVERY.request, .confirm;
- Path discovery: this can be triggered for diagnosis. The goal is to retrieve the detail of each link constituting the forward and reverse route to a given destination node. For more detail, see ITU-T G.9903:2014, 9.4.3.2.4. Associated service primitives used: ADPM-PATH-DISCOVERY.request, .confirm.

The PAN coordinator may also receive from the ITU-T G.9903:2014 modem information about the G3-PLC network. This notification is generated when the adaptation sublayer of a PAN coordinator has received an LBP message from a device on the network indicating that an alternate PAN ID has been detected (see ITU-T G.9903:2014, Table 9-1 for a description of the PAN ID detection handling mechanism). Associated service primitive used: ADPM-NETWORK-STATUS.indication.

For COSEM interface classes used to set up the communication media and to manage it see Clause 8 and IEC 62056-6-2.

6 Identification and addressing schemes

The addressing scheme is described in 5.3.4. This profile does not put any restrictions on client and server SAP addressing.

7 Specific considerations for the application layer services

7.1 Overview

The metering application comprises layers 5 to 7 of the OSI model as well as the application process modelled by the COSEM objects. As described in Figure 2, the model proposed for metering comprises two broad classes of applications: the metering application proper and the applications ensuring the management of the meter.

It is noted that both applications rely natively on UDP, but there is nothing to prevent the future introduction of applications that use TCP.

In the G3-PLC profile, the DLMS/COSEM application strictly complies with the existing standards, namely:

- "COSEM application process" as specified in IEC 62056-6-1 and IEC 62056-6-2;
- "DLMS/COSEM Application layer" as specified in IEC 62056-5-3;
- Security at application level is ensured by the processes specified in IEC 62056-5-3 and IEC 62056-6-2.

Transport of the DLMS/COSEM application protocol by UDP requires the interposition of a "wrapper" in accordance with IEC 62056-4-7:2015. The use of this wrapper allows one to use UDP/IPv6 over ITU-T G.9903:2014.

7.2 Application association establishment and release: ACSE services

Application associations are managed by the COSEM services listed in 7.3.

7.3 DLMS/COSEM services

The DLMS/COSEM services as specified in IEC 62056-5-3 apply, at least the following services shall be supported:

- ACSE services for application association establishment and release:
 - COSEM-OPEN;
 - COSEM-RELEASE;
 - COSEM-ABORT.
- xDLMS services for data exchange:
 - GET;
 - SET;
 - ACTION;
 - EventNotification.

All G3-PLC devices shall use Logical Name (LN) referencing.

7.4 Security mechanisms

Besides the security features provided by IEC 62056-5-3, the MAC layer provides additional security features as specified in ITU-T G.9903:2014.

7.5 Transferring long application messages

Fragmentation and reassembly are provided by the adaptation layer as described in 5.4.

7.6 Media access, bandwidth and timing considerations

There are no special restrictions to be considered.

7.7 Other considerations

7.7.1 UDP DLMS/COSEM wrapper

The UDP DLMS/COSEM wrapper as specified in IEC 62056-4-7:2015 applies, with the selections specified in Table 3. The following conventions apply:

- I = "Informative". The statements of the reference document are provided for information only;
- N = "Normative": The statements of the reference document apply without modifications or remarks;
- S = "Selection": The statements of the reference document apply with the selections specified;
- E = "Extension": The statements of the reference document apply with the extensions specified;
- N/R = "Not Relevant": The statements of the reference document do not apply. An explanation may be given under the part title.

Table 3 – Selections from IEC 62056-4-7:2015

Clause	Title and remarks/modifications	Statement
1	Scope	N
2	Normative references	N
3	Terms, definitions and abbreviations	N
3.1	Terms and definitions	N
3.2	Abbreviations	N
4	Overview	N
5	The DLMS/COSEM connection-less, UDP-based transport layer	N
5.1	General	N
5.2	Service specification for the DLMS/COSEM UDP-based transport layer	N
5.2.1	General	N
5.2.2	The UDP-DATA service	N
5.2.2.1	UDP-DATA.request - Specific UDP ports have been defined in 5.3.5 in order to take full advantage of the compression of the UDP headers by the ITU-T G.9903:2014 Adaptation layer (see ITU-T G.9903:2014, 9.4.2.2 for more details).	S
5.2.2.2	UDP-DATA.indication - The same remarks as for the UDP-DATA.request service primitive apply for the 'Local_UDP_Port' and 'Remote_UDP_Port' parameters.	S
5.2.2.3	UDP-DATA.confirm - The local generation of a UDP-DATA.confirm service primitive does not provide a gain at functional level for the G3-PLC standard, therefore it is ignored.	N/R
5.3	Protocol specification for the DLMS/COSEM UDP-based transport layer	N
5.3.1	General	N
5.3.2	The wrapper protocol data unit (WPDU)	N
5.3.3	The DLMS/COSEM UDP-based transport layer protocol data unit	N
5.3.4	Reserved wrapper port numbers (wPorts)	N
5.3.5	Protocol state machine	N
6	The DLMS/COSEM connection-oriented, TCP-based transport layer	N/R
6.1	General	N/R
6.2	Service specification for the DLMS/COSEM TCP-based transport layer	N/R
6.2.1	General	N/R
6.2.2	The TCP-CONNECT services	N/R
6.2.2.1	TCP-CONNECT.request	N/R
6.2.2.2	TCP-CONNECT.indication	N/R
6.2.2.3	TCP-CONNECT.response	N/R
6.2.2.4	TCP-CONNECT.confirm	N/R
6.2.3	The TCP-DISCONNECT services	N/R
6.2.3.1	TCP-DISCONNECT.request	N/R
6.2.3.2	TCP-DISCONNECT.indication	N/R
6.2.3.3	TCP-DISCONNECT.response	N/R
6.2.3.4	TCP-DISCONNECT.confirm	N/R
6.2.4	The TCP-ABORT service	N/R
6.2.4.1	TCP-ABORT.indication	N/R
6.2.5	The TCP-DATA services	N/R

Clause	Title and remarks/modifications	Statement
6.2.5.1	TCP-DATA.request	N/R
6.2.5.2	TCP-DATA.indication	N/R
6.2.5.3	TCP-DATA.confirm	N/R
6.3	Protocol specification for the DLMS/COSEM TCP-based transport layer	N/R
6.3.1	General	N/R
6.3.2	The wrapper protocol data unit (WPDU)	N/R
6.3.3	The DLMS/COSEM TCP-based transport layer data unit	N/R
6.3.4	Reserved wrapper port numbers	N/R
6.3.5	Definition of the procedures	N/R
6.3.5.1	TCP connection	N/R
6.3.5.2	TCP disconnection	N/R
6.3.5.3	TCP connection abort	N/R
6.3.5.4	Data communication using the TCP-DATA services	N/R
6.3.5.5	High-level state transition diagram of the wrapper sub-layer	N/R
Annex A	Converting OSI-style transport layer services to and from RFC-style TCP function calls	N/R
NOTE References are to IEC 62056-4-7:2015.		

7.7.2 DLMS/COSEM communication profile for UDP/IP networks

The communication profile for UDP/IP networks as specified in IEC 62056-9-7:2013 applies, with the selections specified in Table 4.

Table 4 – Selections from IEC 62056-9-7:2013

Clause	Title and remarks/modifications	Statement
1	Scope	I
2	Normative references	N
3	Terms, definitions and abbreviations	N
4	Targeted communication environments	N
5	Structure of the profile(s) - Only the IPv6 network layer and UDP transport layer are applicable for this standard.	S
6	Identification and addressing scheme - For UDP port definition applicable to this standard, see 5.3.5.	S
7	Supporting layer services and service mapping - Only UDP services are supported by this standard	S
8	Communication profile specific service parameters of the COSEM AL services - Only UDP parameters are supported by this standard	S
9	Specific considerations / constraints	N
9.1	Confirmed and unconfirmed AAs and service invocations, packet types used	N
9.2	Releasing application associations: using RLRQ/RLRE is mandatory	N
9.3	Service parameters of the COSEM-OPEN / -RELEASE / -ABORT services	N
9.4	xDLMS client/server type services	N
9.5	The EventNotification Service and the TriggerEventNotificationSending service	N
9.6	Transporting long messages	N
9.7	Allowing COSEM servers to establish the TCP connection	N/R
9.8	The COSEM TCP-UDP/IP profile and real-world IP networks	N
NOTE References are to IEC 62056-9-7:2013.		

8 Communication configuration and management

The configuration and management processes are specified in 5.5.

9 The COSEM application process

No communication profile specific features/restrictions concerning the COSEM application process are needed.

10 Additional considerations for the use of this profile

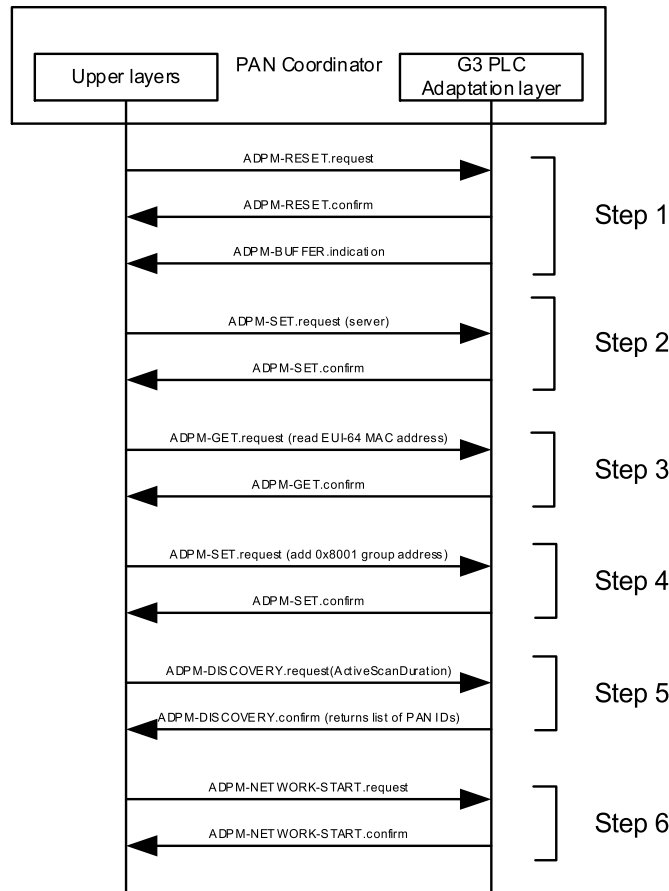
No additional considerations are needed.

Annex A (informative)

Examples

A.1 Example 1: setting up a G3-PLC network dedicated to metering

Before making any G3-PLC communication possible, the data concentrator, typically embedding the PAN coordinator, has to start a G3-PLC network using the mechanisms described in 5.5. The process is shown in Figure A.1:



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Figure A.1 – PAN coordinator initialisation

The following steps are carried out:

- Initialisation – step 1: reset of the G3-PLC modem;
- Initialisation – step 2: configuration of the modem in PAN coordinator mode;
- Initialisation – step 3: the modem's EUI-64 MAC address is read and compared to the address stored within the application layer of the device. If it is different from the modem's parameter, the application layer of the device reconfigures it;
- Initialisation – step 4: the default group address (0x8001) is configured in the G3-PLC modem;
- Initialisation – step 5: a discovery procedure is started as defined in ITU-T G.9903:2014. The duration of the active scan is set to ActiveScanDuration when ADPM-DISCOVERY.request is invoked. The MAC active scan procedure returns a PANDescriptor list;

- Initialisation – step 6: a network start procedure is performed as defined in ITU-T G.9903:2014. By default, the coordinator's short address is set to 0x0000. In addition, a PAN ID is chosen following the rules of the network administrator (out of scope of this document).

A.2 Example 2: smart meters joining a G3-PLC PAN

Meters, also considered as PAN devices according to the G3-PLC terminology, shall initialise their G3-PLC modems and carry out the bootstrapping procedure described below:

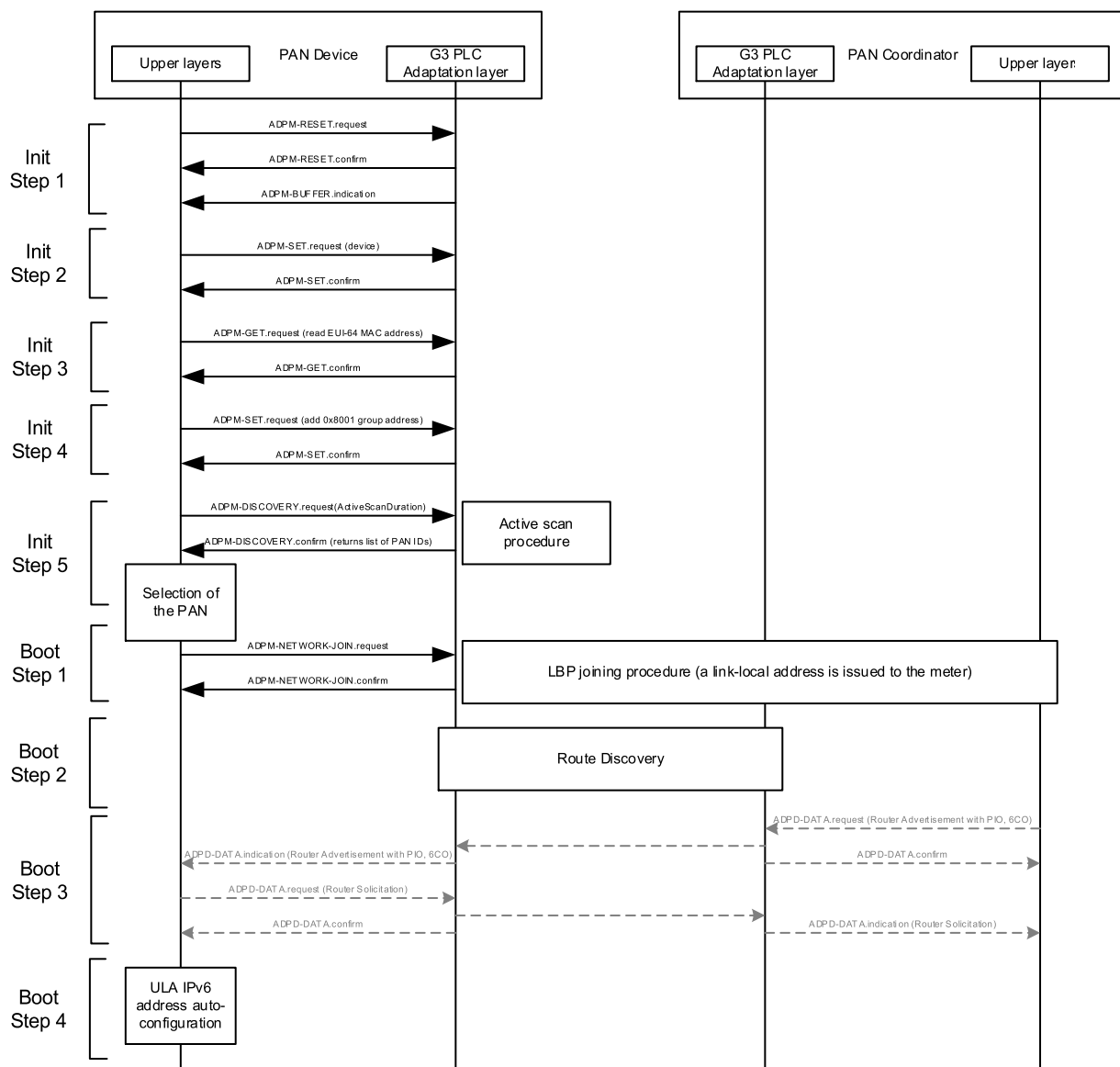
- Initialisation – step 1: reset of the G3-PLC modem;
- Initialisation – step 2: configuration of the modem in PAN device mode;
- Initialisation – step 3: the modem's EUI-64 MAC address is read and compared to the address stored within the application layer of the device. If it is different from the modem's parameter, the application layer of the device reconfigures it;
- Initialisation – step 4: the default group address (0x8001) is configured in the G3-PLC modem;
- Initialisation – step 5: a discovery procedure is started as defined in ITU-T G.9903:2014. The duration of the active scan is set to ActiveScanDuration when ADPM-DISCOVERY.request is invoked. The MAC active scan procedure returns a PANDescriptor list.
- Bootstrapping – step 1: the LowPAN Bootstrapping Protocol procedure is performed as defined in ITU-T G.9903:2014, 9.4.4 in order to join a PAN chosen within PANDescriptor list.
- Bootstrapping – step 2: a route is established between the new PAN device and the PAN coordinator.

A G3-PLC PAN device may be configured using mechanisms such as Neighbour Discovery (IETF RFC 4861) or IPv6 Stateless Address Autoconfiguration (IETF RFC 4862). While Neighbour Discovery configuration allows both ULA and link-local unicast IPv6 addresses to be used, as well as both stateless and stateful multicast addresses, IPv6 Stateless Address Autoconfiguration only allows the use of link-local unicast addresses and stateless multicast addresses.

Consequently, the following steps are performed only if the Neighbour Discovery mechanism is used:

- Bootstrapping – step 3: the PAN device receives a Router Advertisement from the PAN coordinator carrying the context information. Router Solicitation is sent if no Router Advertisement has been received.
- Bootstrapping – step 4: the PAN device autoconfigures its full ULA IPv6 address based on the received context information.

The sequence chart in Figure A.2 illustrates a PAN device's initialisation and complete bootstrapping procedure.



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Figure A.2 – PAN device initialisation and bootstrapping

Annex B (normative)

New COSEM interface classes and OBIS codes

No new COSEM Interface Classes are needed to support this profile.

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